

Interactive Manga Colorization with Fast Flat Coloring

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ABSTRACT

This paper proposes an interactive semi-automatic system for manga colorization. In our system, users can colorize monochrome manga images interactively by scribbling the desired colors. The proposed method creates a high quality colorized image by inputting the original monochrome image after grayscale adjustment and the flat colored image generated from the scribbles to a colorization network. Experiments show that the colorized results yielded by the proposed method are much better than existing methods.

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1 INTRODUCTION

Colorizing Japanese comics (a.k.a. manga) is an important task because while they are traditionally drawn and sold in monochrome format, full-color versions are now attracting a wider audience. To support the manual colorization process, which is time-consuming and costly, a lot of (semi-) automatic methods have been proposed not only for manga but also for illustrations such as [Hensman and Aizawa 2017; Shimizu et al. 2021; Zhang et al. 2018]. Among them, the method proposed in [Shimizu et al. 2021] for manga can output high quality colorization results and can be trained from a small amount of training data by inputting a pair of flat colored and screen tone (i.e., monochrome) images. However, the drawback of [Shimizu et al. 2021] is that the flat colored image has to be manually prepared in advance.

To avoid this problem, we propose a semi-automatic colorization system, where users can interactively colorize screen tone manga images with scribble inputs. The key idea of the proposed system is to use the Watershed algorithm [Vincent and Soille 1991] to rapidly obtain a flat colored image from the scribbles. Even if the quality of the flat colored image is not good, high quality colorized results are obtained by inputting the pair of flat colored and screen tone images to a colorization network.

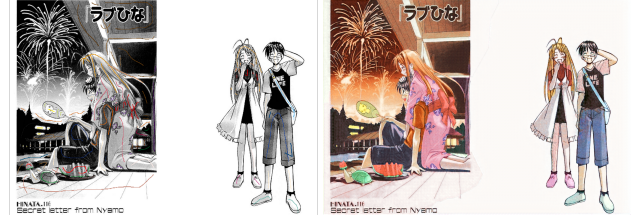


Figure 1: Left: Input manga image and scribbles. Right: Our result. From “LoveHina ©Akamatsu Ken” in Manga109-s dataset [Matsui et al. 2017].

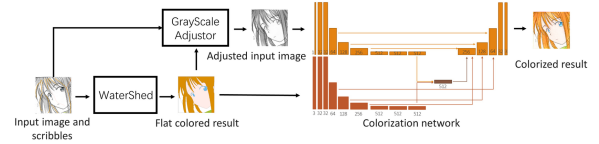


Figure 2: Overall pipeline of the proposed method. ©Akamatsu Ken.

In addition, we propose a pre-processing method (called GrayScale Adjustor: GSA); it adjusts the pixel values of the input screen tone image in order to obtain higher quality results. In manga, screen tone images sometimes have abnormal pixel values (i.e., significantly different gray scale values from the actual color of the objects), which degrades the colorization performance of the network. Such abnormality occurs when the screen tones are used to emphasize particular regions. GSA alleviates this problem by making the pixel values of the screen tone images close to those of flat colored images.

2 PROPOSED METHOD

Fig. 2 shows the overall structure of the proposed method. It consists of three components: (i) flat coloring with Watershed algorithm, (ii) gray scale adjustment, and (iii) colorization by UNet.

2.1 Flat coloring

Given the input screen tone image and K scribbles with the color specified by the user, we first generate the flat colored image by the following procedure.

Formally, i -th scribble ($i = 1, \dots, K$) consists of the user-specified color $\mathbf{c}_i = (r_i, g_i, b_i)$ and the set of N_i pixels $S_i = \{p_1^i, \dots, p_{N_i}^i\}$. We partition the input screen tone image I into K segmented regions

by running the Watershed algorithm¹, where we set the S_i ($i = 1, \dots, K$) as K local minima (i.e., K initial markers). Here, we denote the obtained i -th region as R_i . We obtain the flat colored image I_f by colorizing every pixel in R_i with the color c_i .

2.2 Grayscale Adjustor (GSA)

Before inputting the pair of the flat colored image and screen tone image to the colorization network, we adjust the pixel values of the latter to obtain better results. Let I_g be the gray scale image converted from I_f . As discussed in Sec. 1, the screen tone image I sometimes has significantly different pixel values from I_g (i.e., the actual color of the objects specified by the user) because screen tones can be used for the purpose of emphasis. In order to alleviate this problem, we adjust the pixel values of I so they approach those of I_g in each region.

We first compute the mean difference D_i between I and I_g in each region R_i as follows:

$$D_i = \frac{1}{|R_i|} \sum_{p \in R_i} (I_g(p) - I(p)), \quad (1)$$

where $|R_i|$ is the number of pixels in region R_i , and $I(p)$ and $I_g(p)$ are the values at pixel p in I and I_g , respectively. Then, we adjust the pixel values of I as follows:

$$I'(p) = I(p) + \lambda D_i, \quad p \in R_i, \forall i, \quad (2)$$

where λ is a parameter we set manually. The reason we use the mean difference at each region (not the difference at each pixel) is to keep the local details of the screen tone image.

2.3 Colorization network

After obtaining adjusted screen tone image I' , we input the pair of I' and I_f to the colorization network and get the final colorized result. Similar to [Shimizu et al. 2021], we adopt two-stream UNet structure as shown in Fig. 2. The network training phase uses smooth L1 loss between the output result and ground truth color image.

3 EXPERIMENTS

We trained UNet with training images that were randomly selected from “Nekodama ©Ebifly” in Manga109 dataset [Matsui et al. 2017] and six commercially available comics. The ground truth color images were created by a proficient colorist for “Nekodama ©Ebifly”. For commercially available comics, we used official colorized versions as the ground truth. The total number of training images was 49. We trained UNet over 200 epochs with the learning rate of 0.001. For evaluation, we used five images randomly selected from “Nekodama ©Ebifly” other than the training images. We set λ to 1.

In the evaluation phase, we input line segments (instead of free-hand scribbles) to the proposed method for comparisons with other methods. The color of each line segment was set as the mean of ground truth colors at two edge points of the line segment. The location of each line segment was selected manually.

We compared the proposed method with Two-stage colorization [Zhang et al. 2018] trained on the Danbooru dataset. For this

¹The height of each basin is calculated based on the highest absolute channel difference between the target pixel and its neighbor. For more details, please refer to the watershed function implemented in OpenCV library.

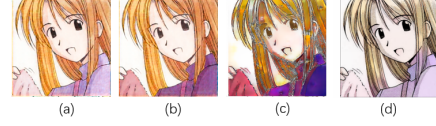


Figure 3: (a) Ours w/o GSA, (b) Ours w/ GSA, (c) [Zhang et al. 2018], and (d) Clip studio paint. ©Akamatsu Ken.

Table 1: PSNR averaged on five test images.

Method	PSNR
Ours w/o GSA	20.59
Ours w/ GSA	21.97
[Zhang et al. 2018]	16.72
Clip studio paint	18.83

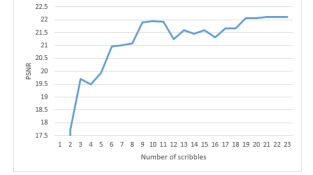


Figure 4: PSNR vs. number of scribbles.

method, we used the same line segments as the proposed method for evaluation. However, because this method takes point hints (not scribbles) as input, we input the center point of each line segment. We also compared the proposed method with Clip studio paint tool, where we manually input line segments as close to those for the proposed method as possible.

Table 1 shows the average PSNR on five test images, and Fig. 3 shows the qualitative comparison on an image from “LoveHina ©Akamatsu Ken”. We observed that the proposed method obtained the best results and that GSA improved its performance. Fig. 4 plots PSNR vs. the number of input scribbles (line segments) for a test image. We observed that about ten scribbles are enough to obtain high quality result.

4 CONCLUSIONS

In this paper, we proposed a semi-automatic colorization method for manga. The proposed method can output high quality results from scribble inputs by creating a flat colored image with the Watershed algorithm and inputting it to the colorization network along with the original screen tone image after grayscale adjustment. The proposed method enables efficient colorization from a small number of input scribbles without manual preparation of flat colored images.

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